

Theory

Conductometry is one of the simplest electrochemical methods of analysis which is based on measurement of electrical conductance of electrolyte solutions. When electric field is applied to the solution, ions experience a force that attracts them towards the oppositely charged electrode. Measurement of conductance afford to determine the concentration of electrolyte through conductometric titration

During titration one of the ions is replaced by other ion and invariably these two ions have different conducting power. If conductance is plotted against the volume of the titrant added, the nature of the curve before the end point will be different from the curve after the end point. On extrapolation of these two curves, they intersect at a point, which corresponds to the end point of the titration. This is the basic principle of conductometric titration

PART A

Conductometric titration

(i) Strong acid and strong base

- i. Rinse the conductivity cell a number of times with conductivity water or double distilled water.
- ii. Pipette out 25 mL of HCl in a beaker and dip the conductivity cell in it, so that the cell should dip completely in solution.
- iii. Note the temperature of the sample solution and accordingly set the temperature control or keep the cell in a thermostat at room temperature.
- iv. Add small amount 0.1 mol dm^{-3} NaOH solution (few drops) from burette, stir it and measure the conductance after each addition.
- v. Take at least five readings beyond the end point.

Plot graph between corrected conductance (G_{corr}) and volume of titrant (NaOH solution). The points of intersection of these lines represent the equivalent point

The increase in volume as the titration goes on should be carefully considered as it affects the linearity of the resulting curve. One of the commonly followed precautions is to correct for dilution using the following equation:

$$G_{\text{corr.}} = G_{\text{obs.}} [(v_1 + v_2)/v_1]$$

where $G_{\text{corr.}}$ and $G_{\text{obs.}}$ are the corrected and observed conductance, respectively; v_1 is the initial volume, and v_2 is the volume of the added reagent.

From the plot, calculate the concentration of HCl solution

(ii) Titration of an acid mixture by NaOH

Repeat the above procedure for the acid mixture of CH_3COOH acid and HCl

From the plot, calculate the concentration of CH_3COOH acid and HCl in the solution

Discussion:

1. Explain the shape of the titration curve obtained for part (i) and (ii)
2. How would the dilution effect as a result of the increase in volume of the solution, when base is added be minimized in the titration?
3. What are the merits of conductometric titration over acid -base titration?

Part B

Determination of solubility product constant of BaSO_4

The conductivity of a saturated solution of a sparingly soluble salt can be determined by subtracting the measured conductivity of the pure solvent from the measured conductivity of the saturated solution, i.e.

$$k_{\text{salt}} = k_{\text{solution}} - k_{\text{water}}$$

The molar conductivity of the salt in saturated solution is given by

$$\Lambda_{\text{salt}} = k_{\text{salt}} / c$$

where c is the concentration. Since the salt is sparingly soluble, its solution can be considered to be infinitely dilute and molar conductivity can be replaced with limited molar conductivity of the salt. That is, limiting molar conductivity Λ° for

any electrolyte can be expressed as the sum of independent contributions from the constituent cations and anions present. This fact is now known as Kohlrausch's Law of the Independent Migration of Ions

$$\Lambda_{salt} = \Lambda_{salt}^{\alpha} = \nu_{+} \lambda_{+}^{\alpha} + \nu_{-} \lambda_{-}^{\alpha}$$

where ν_{+} and ν_{-} the number of cationic and anions in the formula of electrolyte and the limiting ionic conductivities $\lambda_{+}^{\alpha}, \lambda_{-}^{\alpha}$ can be obtained from the literature. The solubility product K_{sp} , for a sparingly soluble salt is given by

$$K_{sp} = C_{+} C_{-}$$

Experimental procedure:

- i. Measure about 0.5g of finely powdered BaSO_4 into a clean, dry watch glass.
- ii. Add to a clean beaker. The watch glass should be washed with distilled water and that washed water should also be added to the beaker.
- iii. To remove impurities, shake the solid with distilled water. Keep aside for a while, for the salt to be collected at the bottom of the beaker. Then the upper layers is decanted and repeat this step for several times.
- iv. An empty reagent bottle should be kept in a water bath at 25°C in order to maintain the temperature of the reagent bottle at 25°C . Add well- washed salt to the reagent bottle and add 50 mL of distilled water. After awhile without disturbing the solid, decant the clear upper solution into a clean beaker and measure the conductivity.
- v. Measure the conductivity of distilled water.
- vi. Determine the solubility product of BaSO_4 using the above equations

* The limiting ionic conductivities of Ba^{2+} and SO_4^{2-} are 110 & 136.6 $\text{S cm}^2 \text{mol}^{-1}$ respectively.